INCH-POUND
MIL-B-1163A
2 September 1988
SUPERSEDING
JAN-B-1163
30 June 1949
(See 6.4)

MILITARY SPECIFICATION

BARS, FLINDERS

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope Flinders bars covered by this specification are intended for use with compasses

1 2 Type Flinders bars covered by this specification shall be of the following type.

Type I

2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications and standards The following specifications and standards form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6605

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STANDARDS

MILITARY

MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.2 Other publications The following document forms a part of this specification to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents These documents also may be available in or through libraries or other informational services)

2 3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3.1 Composition Flinders bars shall be of highest grade commercial pure soft iron and of purity 09 85 percent iron worked in such a manner as to avoid introduction of mechanical strain due to shear or otherwise

3.1.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid

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waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Heat treatment The bars shall be heated slowly and uniformly to a temperature between 1500 and 1600 degrees Fahrenheit (°F) Cooling shall be accomplished in the furnace at a rate not to exceed 350°F per hour until the temperature reaches 1000°F, when cooling shall then proceed at a rate not to exceed 100°F per hour until the temperature reaches 400°F, then cooled in still air to approximately atmospheric temperature

3.3 Dimensions The flinders bars complete shall be 24 inches in length overall and shall be comprised of four sections, each of which shall be cylindrical and 2 inches in diameter throughout One section shall be 12 inches in length one section 6 inches in length, and two sections 3 inches in length

3.4 Case The flinders bar case shall be rectangular in shape, of hardwood, well seasoned and finished, top, sides, and bottom shall be 1/2-inch thick, the sides of the case and shall be dovetailed and secured together by two inside hinges approximately 2 inches from each end, the hinges shall be secured with brass screws, the whole exterior surface of the case shall be varnished, and when the case is closed the top shall be secured to the bottom by a brass hook and screw eye near each end of the case The case shall be approximately 26 inches in length, 5-5/8 inches in width, and 3-1/4 inches deep, all measurements overall The inside of the case shall be lined with green felt and shall be divided into two longitudinal sections, into one of which shall fit the flinders bar In the other there shall be four sections of smooth turned hardwood similar in lengths to the sections of the flinders bar, but of 1-7/8 inches diameter The grain of these wood spacers shall be approximately parallel to the length of the spacer The ends of each longitudinal section of the case inside shall have fitted and secured into them a piece of gray felt about 1/2-inch thick to take up any longitudinal movement or thrust of the bar in transport

3.5 Workmanship Workmanship shall be of highest quality and each section shall have an approved black, lusterless finish

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material

4.2 Sampling

4.2.1 Lot. For the purposes of sampling, a lot shall consist of not more than 1000 flinders bars in cases offered for delivery at one time.

4 2 2 Sampling procedure for lot acceptance inspection For the inspection specified in 4 3, the Naval inspector shall select representative sample bars in cases in accordance with table I

TABLE I Sampling procedure

Number of flinders bars in the lot	Number of samples to be selected
1-20	All
21-50	20
Over 50	30

4 2 3 Sampling procedure for chemical analysis From the iron used in the first lot offered for delivery under each contract or order, and thenceforth in at least one lot chosen from every 10 lots offered for delivery, the Naval inspector shall take a sample for chemical analysis and shall forward it to a Government laboratory.

4 3 Inspection Each of the samples selected in accordance with 4 2.2 shall be subjected to surface inspection by the Naval inspector to determine conformance to the requirements of this specification If any sample flinders bar or case is found to be nonconforming to the requirements of this specification, the entire lot shall be rejected A rejected lot may be resubmitted for Naval inspection, but only after the contractor subsequent to the first Naval inspection, has inspected the lot 100 percent for the deficiency noted and has removed or repaired all nonconforming bars or cases.

4.4 Chemical analysis. The sample selected in accordance with 4.2.3 shall be analyzed to determine conformance to the requirements of 3.1. Acceptance of the first lot shall be withheld pending receipt of a satisfactory test report on the sample representing the lot. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling and inspection of 4 2 2 and 4 3 and acceptance shall not be withheld pending

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receipt of test reports on chemical analyses However, upon receipt of an unsatisfactory test report on a chemical analysis, the Naval inspector shall select a sample from every subsequent lot offered for delivery and shall submit the sample so selected to the Government laboratory for analysis Lots shall then be accepted only upon receipt of a satisfactory report on the samples so selected This additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted

4 5 Inspection of packaging Sample packs, and the inspection of the preservation, packing, and marking for shipment stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition)

5 1 General

5 1 1 Navy shipboard stowage fire-retardant requirements Fire-retardant requirements shall be as specified below

(a) Treated lumber and plywood Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows

Levels A and B - Type II - weather resistant
Category 1 - general use

Level C - Type 1 - non-weather resistant
Category 1 - general use

(b) Fiberboard Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, treated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto

5 2 Level A preservation Bars shall be cleaned, dried and unit protected in accordance with method III requirements of MIL-P-116 In addition, bars shall be protected by a volatile corrosion inhibitor barrier wrap or bag and then placed in unit packs (containers)

5 2.1.1 Unit pack. Unit packs (unit containers) shall conform to carrying case described (see 3.4).

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

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5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be of minimum weight and cube, similar construction, and of uniform size

5.3.2 Level A, B and C containers. Material preserved as specified (see 5.2) shall be packed in shipping containers; plywood, nailed wood, or wirebound boxes, for the level of packing specified (see 6.2), in accordance with MIL-STD-2073-1, appendix C, table VII Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

5.3.2.1.1 Closure Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto

5.3.2.1.2 Weight Wood, plywood, and cleated type containers shall not exceed 150 pounds gross weight

5 3 2 1 3 Waterproofing Unless otherwise specified (see 6 2), level A and when specified (see 6 2), level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186

5 3 3 Commercial Material preserved as specified (see 6 2) shall be packed for shipment in accordance with ASTM D 3951 and herein

5 3 3.1 Container modification Shipping containers exceeding 200 pounds gross weight shall be provided with the minimum of 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base that will support the material and facilitate handling by mechanical handling equipment during shipment

5 4 Marking. In addition to any special marking required (see 6.2), level A, B, and C interior packs and shipping containers shall be marked in accordance with MIL-STD-2073-1, appendix F, and commercial interior packs and shipping containers shall be marked in accordance with ASTM D 3951 In addition bar coding shall be applied in accordance with the marking requirements of MIL-STD-2073-1

6 NOTES

6 1 Intended use Flinders bars covered by this specification are intended for use with compasses

6 2 Ordering data. Acquisition documents should specify the following

(a) Title, number, and date of this specification.
(b) Fire retardant material requirements (see 5.1.1).
(c) Preservation, packing, and marking requirements (see 5.3, 5.3.2, 5 3.3, and 5.4).
(d) When container selection is not at the contractor's option (see 5.3.2).
(e) Caseliner requirements (see 5.3.2.1.3).

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6.3 Subject term (key word) listing.

Chemical analysis
Hardwood
Iron

6.4 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodian
Navy - SH

Review activities
Army - AV
Air Force - 99

Preparing activity
Navy - SH
(Project 6605-0387)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-1163A	BARS, FLINDERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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